

Quantum master equation for coupling mechanism of metamaterial

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ABSTRACT

In this paper, we present a full quantum description for coupling mechanism of the metamaterial. We start the derivation from quantum master equation to fit the spectrum of THz metamaterial. Therefore we can employ the quantum master equation to describe the coupling of metamaterial, such as the electromagnetically induced transparency (EIT) and bound state in the continuum (BIC). In this paper, we build up a bridge between the quantum mechanism and coupling mechanism of metamaterial, so-called quantum–classical analogies. Based on our theory, we can find the more quantum phenomenon with corresponding to the coupling phenomenon of the metamaterial.

Introduction

Quantum–classical analogies [1] is a very active research area, which brings the connection between the quantum theory and classical devices, such as the broadband optical waveguide coupler [2–5], high-efficient wireless energy transfer [6,7], optical and terahertz (THz) half-wave plate [8–10] and broadband surface polaritons plasmons (SPPs) waveguide coupler [11–13]. The idea of quantum–classical analogies is that we can obtain some quantum-like phenomena in classical systems. On the other hand, by inspiration of quantum theory, we can design and solve some issues of the classical devices, such as the narrow-band and low-efficient transfer.

In this paper, we present a full quantum description (quantum master equation) of an open quantum system to fit the spectrum of THz metamaterial with coupling between two metamaterial structures, such as the electromagnetically induced transparency (EIT) and bound state in the continuum (BIC). EIT and BIC originally come from quantum theory. Recently, these ideas have been introduced into the metamaterial and have achieved many results of applications. EIT and BIC spectrums are from the coupling between two metamaterial structures. Normally, EIT in metamaterial has already been explained by coupling of one bright mode and one dark mode [14–16] and BIC can be demonstrated by coupling of two bright modes [17–19]. In this paper, we only present the simulation results of coupling between THz metamaterial structures. However, there are many studies to realize the EIT and BIC based on the coupling between the metamaterial structures in THz frequency range [14,15,20–22], therefore EIT and BIC of THz metamaterial are

very common used in the researches and definitely can be realized by experiment.

Some papers have already mentioned that the metamaterial structures can be considered the quantum energy levels [23–28], however, they do not clearly demonstrate the equivalence of complete quantum theory (based on quantum master equation) and coupling of the metamaterial structures. In this paper, we give the complete quantum theory (quantum master equation) of the coupling phenomena of metamaterial. Furthermore, we exactly illustrate the equivalence of quantum energy levels and metamaterial structures, and propose the complete quantum coupling system of coupling of metamaterial, where our quantum coupling system can be worked for any coupling situations of metamaterial.

The motivations of this paper come from two parts: (i) any physical system and dynamical system should be described by Hamiltonian. Therefore, we believe that Hamiltonian also can describe the THz metamaterial system. In addition, BIC or EIT originally comes from the quantum concept and currently it corresponds to some transmission spectrums of the coupling between two structures of metamaterial. Therefore, our paper demonstrates the connection between coupling of metamaterial and quantum master equation for the first time, which illustrates the fully theory of “quantum–classical analogies” of THz metamaterial. (ii) Furthermore, our method not only works for BIC or EIT based on THz metamaterial, but also works for every coupling pattern of THz metamaterial. Most importantly, this paper can propose

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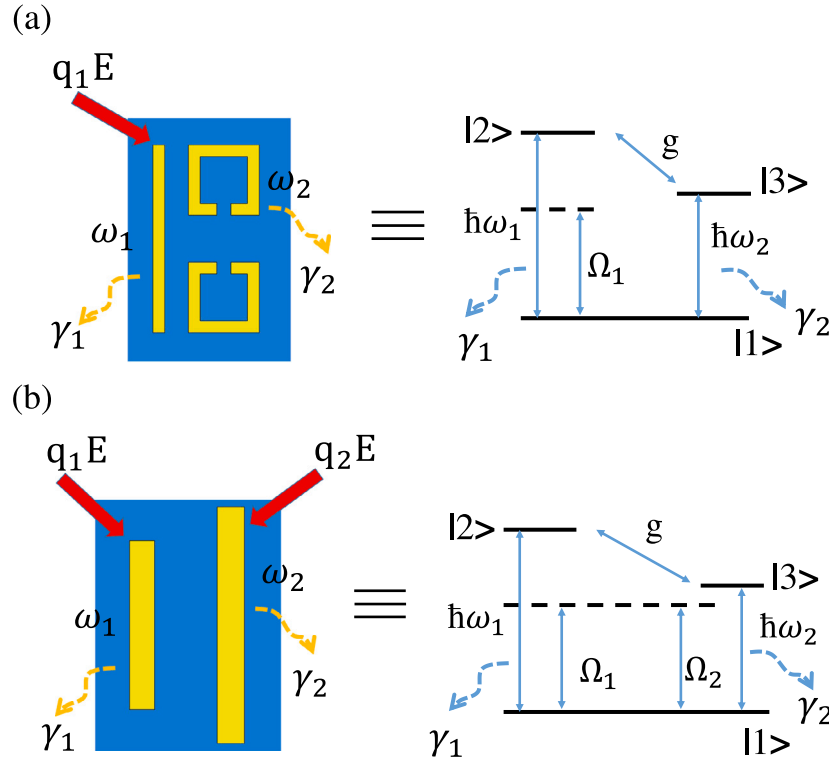


Fig. 1. The schematic figure of coupling between metamaterial structures and corresponding effective energy diagram. (a) Coupling of one bright mode and one dark mode for EIT and (b) coupling of two bright modes for BIC, with corresponding effective energy diagrams respectively.

a new approach to research the coupling phenomena based on metamaterial by employing the quantum master equation. We can obtain some new transmission spectrum corresponding to some ‘quantum’ theory by using our method, not only for BIC and EIT.

Quantum master equation for metamaterial

In this paper, we propose a complete quantum model of the coupling between quantum energy level to be analogue the coupling of metamaterial structures, as shown in Fig. 1. The different metamaterial structures are equivalent to different excited quantum states, where the resonant frequencies ω of different metamaterial structures are equivalent to energy differences of excited states and the ground state with $E = \hbar\omega$ (we take $\hbar = 1$ in this paper). The external field E can excite the different metamaterial structures with the intensities qE (we assume $E = 1$ in this paper), with corresponding the coupling strengths from ground state to excited states Ω . The coupling between different metamaterial structures g is equivalent to the coupling strength between different excited states g .

Thus, we directly write down the Hamiltonian of quantum system in interaction picture as shown in Fig. 1, such as,

$$H = (\omega_1 - \omega_l)|2\rangle\langle 2| + (\omega_2 - \omega_l)|3\rangle\langle 3| - \Omega_1(|2\rangle\langle 1| + |1\rangle\langle 2|) - \Omega_2(|3\rangle\langle 1| + |1\rangle\langle 3|) + g(|3\rangle\langle 2| + |2\rangle\langle 3|), \quad (1)$$

where ω_1 and ω_2 are the resonant frequencies of metamaterial structures 1 and 2. ω_l are the frequency of incident THz wave. Ω_1 and Ω_2 are the intensities of metamaterial structures 1 and 2 by the external field and g is the coupling strength between two metamaterial structures.

Due to the loss of metamaterial structures, it should be written in an open quantum system which can be described as quantum master equation with $\hbar = 1$, such as

$$\frac{d\rho}{dt} = -i[H, \rho]$$

$$+ \gamma_1(2L_{12}\rho L_{12}^\dagger - L_{12}^\dagger L_{12}\rho - \rho L_{12}^\dagger L_{12}) + \gamma_2(2L_{13}\rho L_{13}^\dagger - L_{13}^\dagger L_{13}\rho - \rho L_{13}^\dagger L_{13}), \quad (2)$$

where ρ is the density matrix of quantum states. L_{12} and L_{13} are the Lindblad operators causing by depression from excited states to ground state, where $L_{12} = |1\rangle\langle 2|$ and $L_{13} = |1\rangle\langle 3|$.

Note that we assume that the probability of ground state $|1\rangle$ is close to 1 and probability of excited states $|2\rangle$ and $|3\rangle$ are close to 0 (low excited condition). Therefore, it is easily to obtain $\rho_{11} \gg \rho_{21}, \rho_{12}, \rho_{31}, \rho_{13} \gg \rho_{32}, \rho_{23}, \rho_{11}, \rho_{22}$. Furthermore, this is a classical system, thus $\rho_{12} = \rho_{21}$, $\rho_{13} = \rho_{31}$, $\rho_{32} = \rho_{23}$ to conform to low excited condition. In the numerical calculations, it is very easy to introduce a very large constant C where $\rho_{11} \simeq C^2$, $\rho_{21}, \rho_{31} \simeq C$. We substitute these conditions into quantum master equation (Eq. (2)), then

$$\begin{aligned} i \frac{d\rho_{23}}{dt} &= -\Omega_2\rho_{21}C + \Omega_1\rho_{31}C; \quad (a) \\ i \frac{d\rho_{31}}{dt} &= -\Omega_2 + (\omega_2 - \omega_l)\rho_{31}C + g\rho_{21}C - i\gamma_2\rho_{31}C; \quad (b) \\ i \frac{d\rho_{21}}{dt} &= -\Omega_1 + (\omega_1 - \omega_l)\rho_{21}C + g\rho_{31}C - i\gamma_1\rho_{21}C. \quad (c) \end{aligned} \quad (3)$$

The remarkable papers [28,29] have already shown that the susceptibility $\chi_e(\omega_l) = \langle \rho_{21} \rangle_{ss}$ for the one bright mode and one dark mode (EIT phenomenon) and we can simply expand it into $\chi_e(\omega_l) = \Omega_1 \langle \rho_{21} \rangle_{ss} + \Omega_2 \langle \rho_{31} \rangle_{ss}$ for two bright modes (BIC phenomenon), where $\langle \rho_{21} \rangle_{ss}$ and $\langle \rho_{31} \rangle_{ss}$ are the expected values of ρ_{21} and ρ_{31} in steady state, such that $\rho_{21} = \text{Tr}(|2\rangle\langle 1| \rho_{ss})$ and $\rho_{31} = \text{Tr}(|3\rangle\langle 1| \rho_{ss})$, where ρ_{ss} is the steady state of quantum master equation.

Subsequently, we find the steady state of the quantum master equation where $d\rho_{23}/dt = d\rho_{31}/dt = d\rho_{21}/dt = 0$ and re-write it into matrix form, thus

$$\begin{bmatrix} \omega_1 - \omega_l - i\gamma_1 & g \\ g & \omega_2 - \omega_l - i\gamma_2 \end{bmatrix} \begin{bmatrix} \langle \rho_{21} \rangle_{ss} C \\ \langle \rho_{31} \rangle_{ss} C \end{bmatrix} = \begin{bmatrix} \Omega_1 \\ \Omega_2 \end{bmatrix}, \quad (4)$$

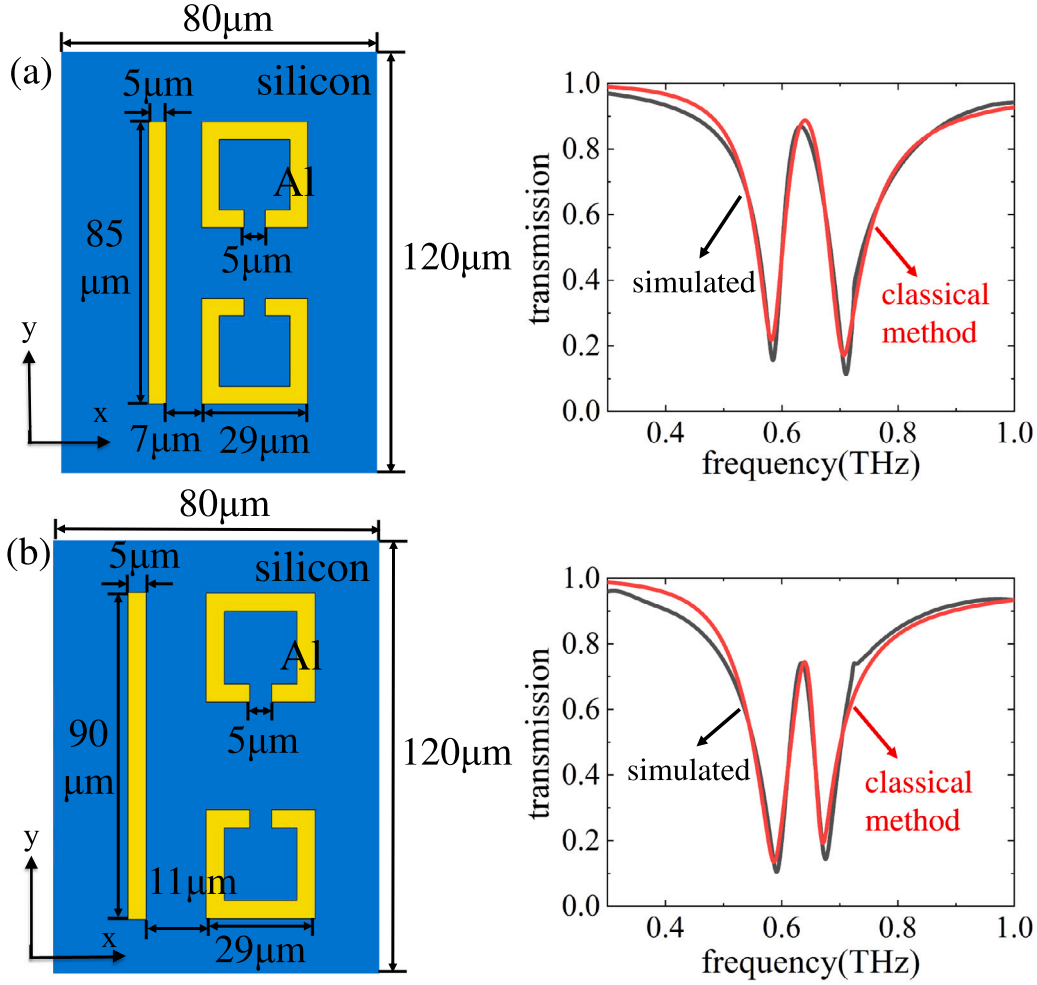


Fig. 2. Two EIT structures and their corresponding spectral spectrum (the black and red curves are simulated and fitting transmission spectrum, respectively).

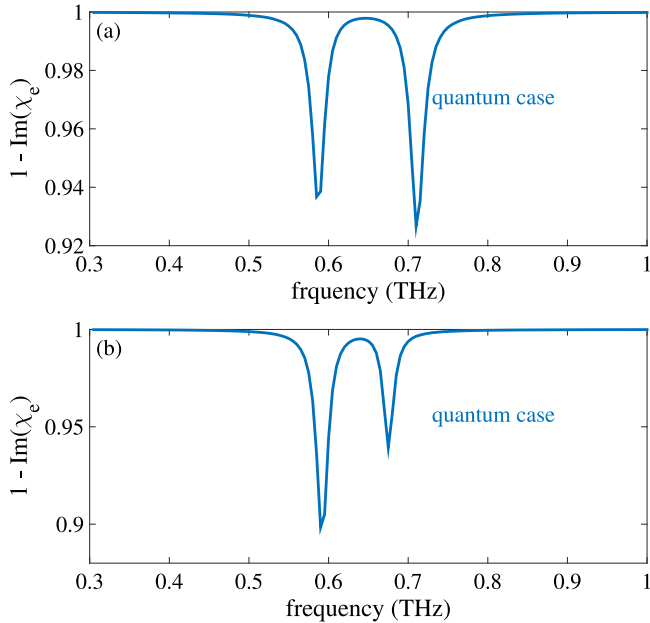


Fig. 3. The transmission spectrum of quantum case with $T = 1 - \text{Im}(\chi_e)$ by numerical solution of reduced quantum master equation as shown in Eq. (3), with corresponding to classical cases 2(a) and (b).

where it is a first order coupled equation of metamaterial structure. Finally, we infer the solution of the $\langle \rho_{21} \rangle_{ss}$ and $\langle \rho_{31} \rangle_{ss}$,

$$\begin{aligned} \langle \rho_{21} \rangle_{ss} &= \frac{((\omega_2 - \omega_l - i\gamma_2)\Omega_1 - g\Omega_2)/C}{(\omega_2 - \omega_l - i\gamma_2)(\omega_1 - \omega_l - i\gamma_1) - g^2}; \\ \langle \rho_{31} \rangle_{ss} &= \frac{((\omega_1 - \omega_l - i\gamma_1)\Omega_1 - g\Omega_2)/C}{(\omega_2 - \omega_l - i\gamma_2)(\omega_1 - \omega_l - i\gamma_1) - g^2}. \end{aligned} \quad (5)$$

The transmission spectrum can be easily given by $T(\omega_l) = 1 - \text{Im}(\chi_e)$.

Numerical calculations

In this section, in order to demonstrate the equivalence between the classical coupled model of metamaterial structures and our quantum system. Firstly, we employ the classical coupled model of metamaterial structures to fit the transmission spectrum of EIT and BIC, to obtain the fitting parameters for Ω_1 , Ω_2 and g . Subsequently, we substitute these parameters into reduced quantum master equation (see Eq. (3)) with low excited condition. Due to the universality of our method, we choose the most widely used structure designs for EIT and BIC as the typical examples. For the EIT case, we choose the coupling between cutting wire (CW) and two split-ring resonators (SRRs), where the CW and SRR work as the bright mode and dark mode respectively [14]. For the BIC case, we give the example of BIC based on two symmetrical cutting wires (CW) and the quasi-BIC with the high Q value based on slightly varying one of CW, which is so-called symmetry-protected BIC [19,30].

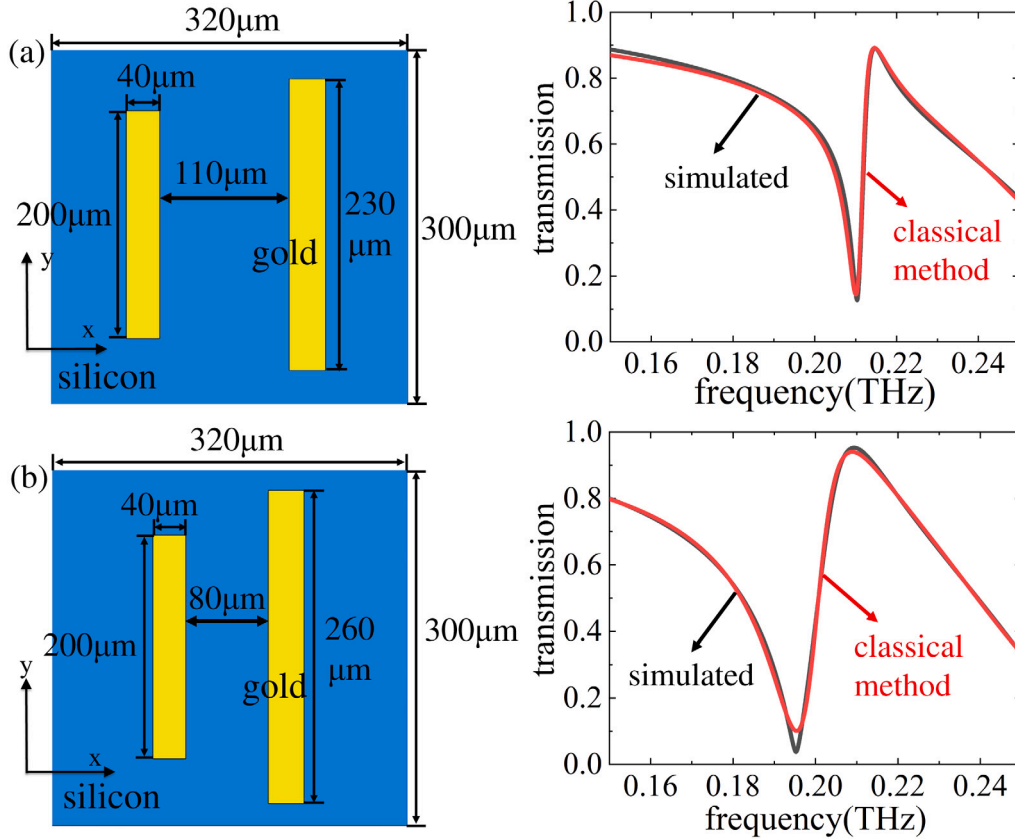


Fig. 4. Two BIC structures and their corresponding spectral spectrum (the black curve and red curve are simulated and fitting transmission spectrum, respectively).

Electromagnetically induced transparency (EIT)

The EIT of metamaterial can be demonstrated from the coupling between one bright mode and one dark mode of two metamaterial structures and we take the mode notations of two metamaterial structures as a and b . The first order classical coupled equation of metamaterial structures, such as

$$\begin{bmatrix} \omega_l - \omega_1 - i\gamma_1 & g \\ g & \omega_l - \omega_2 - i\gamma_2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} q_1 E \\ 0 \end{bmatrix}, \quad (6)$$

where ω_1 and ω_2 are the resonant frequencies of metamaterial structures and ω_l is the frequency of incident THz wave. γ_1 and γ_2 are the loss of metamaterial structures. q_1 is the coupling strength excited bright mode a by external field and g is the coupling strength between two metamaterial structures.

$$\begin{aligned} a &= \frac{q_1(\omega_l - \omega_2 - i\gamma_2)E}{(\omega_l - \omega_1 - i\gamma_1)(\omega_l - \omega_2 - i\gamma_2) - g^2}; \\ b &= \frac{-q_1 g E}{(\omega_l - \omega_1 - i\gamma_1)(\omega_l - \omega_2 - i\gamma_2) - g^2}. \end{aligned} \quad (7)$$

The classical transmission spectrum becomes

$$T = \left| \frac{c(1+n)}{c(1+n) - i\omega_l \chi_e} \right|, \quad (8)$$

where the classical susceptibility $\chi_e = q_1 a$. c is the light velocity in vacuum and $n = 3.88$ is the refractive index of the substrate.

We takes two examples of EIT phenomena based on a typical structure [14]. In the result of Fig. 2(a), we can obtain resonance frequency $\omega_1 = 0.6493$ THz, $\omega_2 = 0.6391$ THz, the loss $\gamma_1 = 0.01293$ THz, $\gamma_2 = 0.006193$ THz, $q_1 = 1.35$ and coupling strength $g = 0.06156$ THz. Fig. 2(b) is an example of the second EIT with the parameters $\omega_1 = 0.6146$ THz, $\omega_2 = 0.6425$ THz, the loss $\gamma_1 = 0.009385$ THz, $\gamma_2 = 0.006537$ THz, $q_1 = 1.418$ and coupling strength $g = 0.0391$ THz.

The metamaterial EIT is equivalent to the quantum energy system as shown in Fig. 1(a). Subsequently, we substitute those parameters into the reduced quantum master equation (Eq. (3)) with the low excited condition. The numerical calculations of quantum master equation are demonstrated in Fig. 3(a) and (b) which are corresponding to metamaterial EIT as shown in Fig. 2(a) and (b).

Bound states in the continuum (BIC)

The BIC of metamaterial can be described by the two bright modes coupling, and we make two structures as a and b . The first-order coupling mode equation is

$$\begin{bmatrix} \omega_l - \omega_1 - i\gamma_1 & g \\ g & \omega_l - \omega_2 - i\gamma_2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} q_1 E \\ q_2 E \end{bmatrix}, \quad (9)$$

Subsequently, we can obtain the energy amplitudes a , b of each metamaterial structure by solving the Eq. (1), as shown

$$\begin{aligned} a &= \frac{q_1(\omega_l - \omega_2 - i\gamma_2)E - q_2 g E}{(\omega_l - \omega_1 - i\gamma_1)(\omega_l - \omega_2 - i\gamma_2) - g^2}; \\ b &= \frac{q_2(\omega_l - \omega_1 - i\gamma_1)E - q_1 g E}{(\omega_l - \omega_1 - i\gamma_1)(\omega_l - \omega_2 - i\gamma_2) - g^2}. \end{aligned} \quad (10)$$

Then, we can calculation the transmission equation, such as $T = 1 - \text{Im}(\chi_e)$, where the classical susceptibility $\chi_e = q_1 a + q_2 b$ and then we also utilize two example of BIC phenomena based on a typical and simple structure [18,19]. For the first example (Fig. 3(b)), We can use the transmission formula to fit the simulation curve and get resonance frequency $\omega_1 = 0.2866$ THz, $\omega_2 = 0.2133$ THz, the loss $\gamma_1 = 0.05208$ THz, $\gamma_2 = 0.0007096$ THz, $q_1 = 0.2252$, $q_2 = 0.003573$ THz and coupling strength $g = 0.01501$ THz. Another example gets parameters: $\omega_1 = 0.2579$ THz, $\omega_2 = 0.2068$ THz, $\gamma_1 = 0.05195$ THz, $\gamma_2 = 0.00109$ THz, $q_1 = 0.2104$, $q_2 = 0.0076$ THz and $g = 0.02649$ THz.

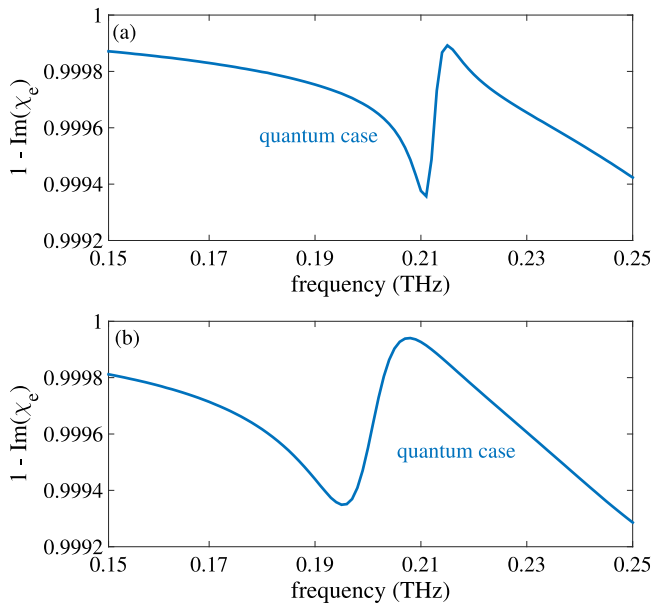


Fig. 5. The transmission spectrum of quantum case with $T = 1 - \text{Im}(\chi_e)$ by numerical solution of reduced quantum master equation as shown in Eq. (3), with corresponding to classical cases 4(a) and (b).

The metamaterial BIC is equivalent to the quantum energy system as shown in Fig. 1(b). Subsequently, we follow the same steps above to obtain the numerical calculations of quantum master equation are demonstrate in Fig. 5(a) and (b) which correspond to metamaterial EIT as shown in Fig. 4(a) and (b).

As we can obtain from the quantum results (see Figs. 3 and 5), the evolution of quantum system have the same spectrums with corresponding to classical spectrums (as shown in Figs. 2 and 4) by using the classical fitting parameters. Therefore, we can demonstrate that our quantum energy diagram is equivalent to the classical models. The significance of this work is not only suitable for coupling of one bright mode and one dark mode or two bright modes, but also work for coupling of arbitrary bright modes and dark modes. Therefore, every combination of coupling of bright modes and dark modes can obtain the corresponding quantum picture. Furthermore, we can construct the quantum energy diagram with corresponding coupling of metamaterial to obtain the new 'quantum phenomenon' with corresponding metamaterial system based on quantum master equation.

Discussion

For the simulations of EIT and BIC of THz metamaterial, we obtain the transmission spectrums of EIT and BIC by using *CST Microwave Studio*TM. We set the polarization direction of terahertz wave as y direction, the boundary conditions of x and y directions as Unitcell, Zmin direction as open, and Zmax as open (add space). In addition, we set the conductivity of aluminum as $3.56 \times 10^7 \text{ S/m}$ and the permittivity of silicon as 11.9. We employ the constant conductivity in the simulation during our simulation. Drude conductivity is more accurate in the simulations, however, using the constant conductivity is good enough for *CST Microwave Studio*TM.

For the fitting illustration, Firstly, we can read out the resonant frequencies ω and losses γ . For the EIT case, parameters ω_1 and γ_1 in the coupled mode equation (see Eq. (6)) can be obtained by simulating the bright structure with y polarization. We set the polarization direction of the terahertz wave as x direction to obtain transmission spectrum which give the parameters ω_2 and γ_2 . For BIC, parameters ω_1 , γ_1 , ω_2 and γ_2 in the coupled mode equation (see Eq. (9)) can be separately obtained by simulating the transmission spectrums of two bright modes,

respectively. The loss in the metal of metamaterial structures can affect the loss rate γ_1 and γ_2 of the coupled mode equations. In the simulations of EIT or BIC, we separately simulate the transmission spectrum of two metamaterial structures to obtain full width at half maxima (FWHM) of the spectrum which are the loss rates of γ_1 and γ_2 respectively. Secondly, we can employ the least square method algorithm to fit the best values of coupling strengths g .

Conclusion

In this paper, we obtain the classical coupling of metamaterial structures with corresponding equivalent quantum energy diagram. We demonstrate the equivalence of our quantum model and classical coupling of metamaterial structures from theoretical derivation and numerical calculations of quantum master equation. We open a new approach of the quantum analog to classical coupling of metamaterial structures.

CRediT authorship contribution statement

Wei Huang: Conceptualization, Methodology, Software, Writing – original draft. **Songyi Liu:** Writing – review & editing, Software. **Shiting Cao:** Writing – review. **Chu Guo:** Methodology. **Shan Yin:** Data curation, Editing. **Xinwei Du:** Writing – review, Supervision. **Wentao Zhang:** Funding acquisition, Support, Supervision.

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